

выполним эквивалентное преобразование схемы и определим calculные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 500 = 3141.5927 \text{ rad/s}$$

$$E_1 = \frac{E_{m1}}{\sqrt{2}} \angle \varphi_1 = 106.066 \angle 28^\circ \text{ V}$$

$$J_2 = \frac{J_{m2}}{\sqrt{2}} \angle \varphi_2 = 56.569 \angle 80^\circ \text{ A}$$

$$E_2 = \frac{J_2}{G_{m2}} = \frac{28.28 + j48.92}{0.3} = 94.28 + j163.3 \text{ V}, R_{m2} = \frac{1}{G_{m2}} = \frac{1}{0.3} = 3.333 \Omega$$

$$E_{eq} = E_1 - E_2 = 93.65 + j49.79 - (94.28 + j163.3) = -0.63 - j113.5 \text{ V}$$

$$Z_{eq, m1} = (R_{m1} + R_{m2}) + jX_{Lm1} = (5 + 3.333) + j2 = 8.33 + j2 \Omega = 8.57 \angle 13.5 \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 3141.5927 \cdot 0.0012 = 3.77 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{3141.5927 \cdot 0.00005} = 6366 \Omega$$

$$X_{L2} = \omega L_2 = 3141.5927 \cdot 0 = 0 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{3141.5927 \cdot 0.00003} = 1061 \Omega$$

$$X_{L3} = \omega L_3 = 3141.5927 \cdot 0.025 = 78.54 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{3141.5927 \cdot 0.000005} = 63662 \Omega$$

$$Z_1 = R_1 + jX_{L1} - jX_{C1} = 6 + j3.77 - j6366 = 6 - j26$$

$$Z_2 = R_2 + jX_{L2} - jX_{C2} = 25 + j0 - j1061 = 25 - j1061$$

$$Z_3 = R_3 + jX_{L3} - jX_{C3} = 32 + j78.54 - j63662 = 32 + j1488$$

Находим эквивалентное сопротивление нагрузки и входное сопротивление цепи

$$Z_{23} = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(25 - j1061)(32 + j1488)}{57 + j427} = 16.75 - j0.69$$

$$Z_{eq, load} = Z_1 + Z_{23} = 6 - j26 + (16.75 - j0.69) = 22.75 - j32.8$$

$$Z_{in} = Z_{eq, m1} + Z_{eq, load} = 8.33 + j2 + (22.75 - j32.8) = 31.09 - j12.8$$

Определим значения токов

$$I_1 = \frac{E_{eq}}{Z_{in}} = \frac{-0.63 - j113.5}{31.09 - j12.8} = 0.13 - j3.65 \text{ A}$$

$$I_2 = I_1 \frac{Z_3}{Z_2 + Z_3} = 0.13 - j3.65 \cdot \frac{32 + j1488}{57 + j427} = 0.87 - j2.08 \text{ A}$$

$$I_3 = I_1 - I_2 = 0.13 - j3.65 - (0.87 - j2.08) = -0.74 - j1.57 \text{ A}$$

Выполняем проверку расчетов по закону Кирхгофа

$$\Delta I = 0 + j0, \Delta U_1 = 0 + j0, \Delta U_2 = 0 + j0$$

$$\varepsilon_{KCL} = 0\%, \varepsilon_{KVL} = 0\%$$